Return Visits & Contacts App — Master Implementation Guide (V5)

Current user workflows, privacy model, PWA behavior, and deployment guide

Updated 27 October 2026 | Defines the City → Region → Area hierarchy

Contents

• Product overview and boundaries
• Architecture and local data protection
• First use and settings
• Contact and visit-log workflows
• Location, map, and urgency
• Backups and restore
• Sharing one contact
• Install, offline behavior, and deployment
• City, region, and area hierarchy

1. Product Overview and Boundaries

Return Visits is a free-to-use client-side Progressive Web App with no app fees or subscription. It is intentionally a simple tool for maintaining contact details and tracking return visits, not a full-featured contact-management app, and may have fewer features or be less reliable than paid products. It is provided as-is. Internet/mobile data and third-party services remain subject to their providers' charges and terms. The app keeps contact details, visit notes, and map locations on the user's device. Each browser/device has its own independent dataset. There are no user accounts, server-side contact records, or automatic cross-device synchronization.
The app is published from the public grasfulls/return-visits-app GitHub repository using GitHub Pages. Hosting and third-party service availability are subject to their providers’ current terms; do not promise that any external service will remain free or available indefinitely.

2. Architecture and Local Data Protection

The app uses static HTML, CSS, and JavaScript. Contact and settings values are encrypted before being written to IndexedDB. Browser LocalStorage is not the main data store; it is used for limited state such as the backup reminder timestamp and compatibility with legacy migration. Encryption uses AES-GCM with a 256-bit key. The key is derived from the passphrase with PBKDF2-SHA-256 (310,000 iterations) and a random salt. Each encrypted value uses a fresh

random IV. The key is held in page memory for the current session, not saved in browser storage.
A passphrase is optional and is off by default; without one, the app opens immediately and stores settings and contacts unencrypted. A user can turn on a passphrase in Settings by choosing and confirming a passphrase of at least eight characters; the existing data is encrypted and verified before the unencrypted copies are replaced. Once a passphrase is on, it is required each time the app opens. A user can then change it by entering the existing passphrase and confirming a new one of at least eight characters. The settings and contacts are decrypted, re-encrypted with fresh encryption metadata, verified, and replaced atomically. Other browser/Home Screen copies and already-exported backups are not changed. There is no passphrase recovery; losing the current passphrase before changing it makes that copy's encrypted records unreadable.

3. First Use and Settings

No passphrase is required on first launch; the app opens directly. To protect the data, open Settings, choose Turn On Passphrase, and enter and confirm a passphrase of at least eight characters. On later launches, enter that passphrase to unlock the database. To change it, open Settings, enter the current passphrase, and enter and confirm a new passphrase. Existing exported backups continue to use the passphrase with which each backup was created. To stop using a passphrase, choose Turn Off Passphrase in Settings and enter the current passphrase once; the data is then stored unencrypted and the app opens without an unlock screen.
Settings contains city, region, and area lists. Each region belongs to one city, and each area belongs to one region. Users can add a region under the selected city from Settings or directly while adding a contact; areas can be added under a region. Edit renames a city, region, or area and updates existing references, including the default city where applicable. Deletion is blocked while any contact or child location uses the entry; those assignments or children must be reassigned or removed first. The app does not provide a multi-state master list or location-based regional onboarding.

4. Contact and Visit-Log Workflows

A contact form can create an Area directly from the Area selector, even when existing areas are listed; the new area is saved beneath the selected City and Region and selected for that contact. A contact includes a stable unique ID, creation time, first name (required), optional last name, email, phone, street-address text, optional city, state, optional region and area, optional latitude and longitude, and visit history. Email and phone are optional; U.S. phone numbers use the format (555) 123-4567. A contact can have at most one city, one region, and one area. Region requires a city; area requires a region. Location fields can be completed or edited later without changing the contact ID. Different contacts may share the same name.
A visit-log entry stores an ID, timestamp, display date, and note text. Logs display newest first and can be edited or deleted. New logs can be added from the contact form or quickly from a contact card. The card provides edit, encrypted export, print, share, navigate, and delete

actions. Navigate opens Google Maps directions using saved coordinates when available, or the contact address otherwise. Share prepares a plain-text message with available contact details and a directions link for Text / SMS, WhatsApp, or Email; visit notes are excluded. This text is not encrypted by the app; use encrypted contact export for app-to-app transfer.
The contact list supports text search over names, email, phone, address, visit-note text, and assigned areas; urgency, city, region, and area filters; and newest-first, oldest-first, and first-name alphabetical sorting. City, region, and area filters follow the same hierarchy. Print Visible Contacts prints exactly the contacts shown by the current search and filters, confirms the count, groups each contact once under their most specific location, and includes available phone and email details plus only the latest visit note. The printer-friendly layout omits app controls.
In Edit Contact, Print opens a note choice: Most recent note only (the default) or All notes. Printing is disabled while contact edits are unsaved; save first to print the updated record. The browser/device print dialog can send the layout to a printer or save it as PDF when supported.

5. Location, Map, and Urgency

Map coordinates are optional. In the contact form, the user can choose Use Device Location to request device GPS or Find Address or City Pin to search the entered street address, city, and state. GPS requests high accuracy with no cached location, but actual accuracy varies and is reported when available. If a partial street address has multiple matches, the user chooses the correct result; the matching full street name is filled in and a pin is placed. If no full-address match is found, the app also searches by the street name without its leading house number. A city-only search places an approximate city pin. The user can also search a nearby cross-street or intersection, or refine pins by tapping the contact form map, dragging the pin, or editing coordinates. Pins can be dragged on the Map view, where their new coordinates save automatically. Contact creation and editing do not request location automatically; contacts without coordinates have no map pin. Whenever the Map is open with no visible contact pins, it centers approximately on the default city and state in Settings and sends that city and state to Nominatim for the lookup. Street addresses and newly entered location names are capitalized when the user leaves the field or saves; recognized U.S. state names are stored as standard two-letter abbreviations, and state codes are uppercased. Street suffixes are displayed in abbreviated form throughout the app, including for previously saved contacts (for example, Avenue as Av, Road as Rd, Court as Ct, and Lane as Ln).
Address, partial street-name, and city searches are sent to OpenStreetMap Nominatim only after the user selects Find Address or City Pin. A cross-street or intersection query is sent only after the user selects Search Near Cross-Street. Whenever the Map is open with no visible contact pins, the default city and state are sent to Nominatim to center the map approximately. Device location requires browser permission. Map tiles and externally hosted Leaflet assets require an internet connection.
Urgency is calculated from the latest visit-log date, or from the contact creation date if there are no logs: Normal is 0–7 days, Urgent is 8–14 days, and Immediate is 15 or more days. The map pin colors reflect these categories: green, yellow, and red.
The map displays only contacts with valid coordinates that match the current search and filters in the contact-list header (text, urgency, city, region, and area). Changing any of these filters

updates both the contact list and map. There are no separate map-only filters. Each matching pin opens a summary popup with an Edit Details action. Pin clustering is not implemented.
The contact modal keeps its Save and Exit actions in a sticky footer while the form area scrolls. Its sizing accounts for mobile viewport and safe-area constraints.

6. Full-Data Backup and Restore

Choose Settings → Export Data Backup to download a JSON backup of settings and all contacts. If a passphrase is on, the backup is encrypted and requires its passphrase to restore; otherwise it is an unencrypted backup. The browser chooses the download destination; the app cannot select a folder. On iPhone, check Files → Downloads; elsewhere, check the browser's Downloads location.
Restore can be started from Settings or, when a passphrase is on, from the unlock screen on a new device. For an encrypted backup, enter the backup passphrase; unencrypted backups need none. Confirm replacement when prompted. Restore replaces the current local dataset; on a fresh device that uses a passphrase, the restored backup passphrase becomes the passphrase used to unlock that data.
Older unencrypted full-app backup files are supported with a warning and are encrypted when imported. Treat any legacy plaintext backup as sensitive and delete or securely store it after migration.
The monthly reminder begins without an immediate first-run prompt and is shown only when the device contains at least one contact. It is a reminder to open Settings and export; it does not automatically create or upload backups.

7. Sharing One Contact

Select the download icon on a contact card to export that one contact, including visit history, as an encrypted contact file. Set a separate sharing passphrase of at least eight characters and confirm it. Give the recipient the passphrase through a different channel from the file.
The recipient can use the regular website or install the PWA. In Settings → Share a Contact, choose the encrypted file, enter its sharing passphrase, and confirm import. Import adds a new contact without replacing existing contacts, assigns a local ID, and stores it using the recipient device's settings (encrypted if the recipient has turned on a passphrase).

8. Install, Offline Behavior, and Deployment

The app can be used in a regular browser; PWA installation is optional. For the best iPhone Home Screen app experience, open the HTTPS site in Safari, use Share, then Add to Home Screen. If the link opened inside another app, use its menu to open it in Safari; a website cannot force the browser. On Android, open the site in Chrome and use Install app or Add to Home Screen. Browser and Home Screen copies may store separate data, so set up the preferred copy before entering contacts or back up before switching.

The service worker caches the same-origin app shell and uses a network-first strategy with cached responses as a fallback. This does not provide offline map tiles, address search, or guaranteed first-load access to externally hosted Leaflet assets.
For a release, commit and push changes to the GitHub repository and wait for the GitHub Pages deployment workflow to succeed. Refresh the site after deployment; installed PWAs update their cached app shell when the new files are fetched online.

9. City, Region, and Area Hierarchy

Each region belongs to one city, and each area belongs to one region. North, South, East, and West are initially available as region names under each existing city. Users may add regions under a selected city and freely named areas under a selected region. Names may be reused in different parent locations.
A contact may have at most one city, one region, and one area. Region and area are optional; the form enables Region only after City is selected and Area only after Region is selected. This permits quick initial entry while preventing contradictory parent-child location assignments.
On upgrade, contacts with multiple area assignments keep the first listed area; additional assignments are removed from that contact. The contact ID, visit history, and other details remain unchanged. Legacy area-list strings are migrated under an Unassigned region so users can review and reassign their location later.